

MATH 3260 - Chapter 2 Notes: Matrices

Feedback-Revised Student Notes

Kennesaw State University - MATH 3260

Fall 2026 revision

Contents

Chapter overview	2
Section 2.1 - Operations with Matrices	2
Learning objectives	2
Addition, subtraction, and scalar multiplication	3
Matrix multiplication	3
Matrix equations and column combinations	4
Partitioned matrices	5
Section 2.1 terminology	5
Section 2.2 - Properties of Matrix Operations	6
Learning objectives	6
Algebra of matrix addition and scalar multiplication	6
Properties of matrix multiplication	6
Identity and zero matrices	8
Transpose	8
Section 2.2 terminology	9
Section 2.3 - The Inverse of a Matrix	10
Learning objectives	10
The 2×2 formula	10
Gauss-Jordan method for a 3×3 inverse	11
Singular matrices	12
Properties of inverses	12
Solving a system with an inverse	12
Section 2.3 terminology	13
Section 2.4 - Elementary Matrices and LU-Factorization	14
Learning objectives	14
Bridge: elementary matrices lead to LU	14
The three types of elementary matrices	14
Factoring a matrix into elementary matrices	15
LU-factorization	15
Solving a system with LU	17
Section 2.4 terminology	18
Section 2.5 - Markov Chains	19
Learning objectives	19
Stochastic matrices	19
Steady state	20
Three-state travel-mode chain	20
Absorbing Markov chain: hiking outcomes	20
Section 2.5 terminology	21
Section 2.6 - More Applications of Matrix Operations	22
Learning objectives	22
Matrix cryptography	22

Leontief input-output model	23
Least squares regression	24
Section 2.6 terminology	25

Chapter 2 Summary	26
Additional clarification: multiple right-hand sides	26

These notes are synchronized with the feedback-revised student notebook. They emphasize definitions, theorems, worked mathematics, conceptual connections, and interpretation while omitting notebook-specific code and worksheet instructions.

Chapter overview

1. **Section 2.1:** Operations with Matrices
2. **Section 2.2:** Properties of Matrix Operations
3. **Section 2.3:** The Inverse of a Matrix
4. **Section 2.4:** Elementary Matrices and LU-Factorization
5. **Section 2.5:** Markov Chains
6. **Section 2.6:** More Applications of Matrix Operations

Each section ends with a terminology table, a worksheet, and complete instructor solutions.

Example policy

All worked examples and numerical values in these notes are original. Personal-interest themes are used selectively: Spurs, Backstreet Boys, hiking, badminton, driving, travel, and R&B music. Any prices, transition probabilities, or datasets are fictional teaching models unless explicitly stated otherwise.

Section 2.1 - Operations with Matrices

Learning objectives

By the end of this section, students should be able to:

- identify matrix size and individual entries;
- determine whether two matrices are equal;
- add, subtract, and scale matrices;
- determine whether a matrix product is defined and find its size;
- compute matrix products by row-column multiplication;
- write a system as $A\mathbf{x} = \mathbf{b}$;
- interpret $A\mathbf{x}$ as a linear combination of the columns of A ;
- understand simple block or partitioned matrices.

Definition: matrix and size

An *mimesn* matrix is a rectangular array with m rows and n columns. The entry in row i , column j is denoted by a_{ij} . Two matrices are equal only when they have the same size and every pair of corresponding entries is equal.

Original example: equality of matrices

Let

$$A = \begin{bmatrix} 9 & 20 \\ 21 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 9 & p \\ 21 & 1 \end{bmatrix}.$$

The matrices have the same size. They are equal exactly when corresponding entries agree, so

$$p = 20.$$

A 2×1 matrix is a **column vector**, and a 1×2 matrix is a **row vector**.

Addition, subtraction, and scalar multiplication

Matrices can be added or subtracted only when their sizes agree. Operations are performed entry by entry:

$$A + B = [a_{ij} + b_{ij}], \quad cA = [ca_{ij}].$$

Original application: fictional Spurs fan-shop inventory

Suppose two kiosks record quantities of caps, shirts, and posters during two time blocks:

$$K_1 = \begin{bmatrix} 9 & 20 & 21 \\ 12 & 18 & 15 \end{bmatrix}, \quad K_2 = \begin{bmatrix} 11 & 16 & 14 \\ 8 & 21 & 20 \end{bmatrix}.$$

Rows represent kiosks and columns represent the three products. Then

$$K_1 + K_2 = \begin{bmatrix} 20 & 36 & 35 \\ 20 & 39 & 35 \end{bmatrix}.$$

If the second time block is weighted twice in a planning score, then

$$K_1 + 2K_2 = \begin{bmatrix} 31 & 52 & 49 \\ 28 & 60 & 55 \end{bmatrix}.$$

Common mistake

Matrix addition is not based on row-column products. Addition is entry-by-entry. Matrix multiplication, introduced next, uses row-column dot products.

Matrix multiplication

If A is $m \times n$ and B is $n \times p$, then AB is defined and has size $m \times p$. The inside dimensions must match:

$$(m \times n)(n \times p) = m \times p.$$

The (i, j) entry is the dot product of row i of A and column j of B :

$$(AB)_{ij} = \sum_{k=1}^n a_{ik}b_{kj}.$$

Original application: badminton training plans

Two players complete three drill types. Let

$$D = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

record drill counts. Let

$$S = \begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 1 & 3 \end{bmatrix}$$

record two scores per drill. Then D is 2×3 , S is 3×2 , and DS is 2×2 .

For the upper-left entry,

$$(DS)_{11} = 1(1) + 2(2) + 0(1) = 5.$$

For the upper-right entry,

$$(DS)_{12} = 1(0) + 2(1) + 0(3) = 2.$$

Continuing gives

$$DS = \begin{bmatrix} 5 & 2 \\ 3 & 4 \end{bmatrix}.$$

Reading the multiplication visual

The (i, j) entry of AB is obtained by taking **row i of A** dotted with **column j of B** . Therefore a “row 1, column 1” annotation is identifying the two pieces used to form $(AB)_{11}$; the arrow is only a visual guide.

Matrix equations and column combinations

Consider

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & 0 \\ 0 & 1 & 2 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 9 \\ 20 \\ 21 \end{bmatrix}.$$

Then

$$A\mathbf{x} = 9 \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} + 20 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + 21 \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 50 \\ 38 \\ 62 \end{bmatrix}.$$

This illustrates a central interpretation:

$A\mathbf{x}$ is a linear combination of the columns of A , using the entries of $\mathbf{x}$ as coefficients.

Partitioned matrices

A matrix can be divided into blocks when the partitions are compatible. For example,

$$A = \left[\begin{array}{cc|c} 1 & 1 & 1 \\ 2 & 1 & 0 \\ 0 & 1 & 2 \end{array} \right] = [A_1 \quad A_2],$$

where A_1 contains the first two columns and A_2 contains the third. Block notation is useful when a matrix has a natural structure or when several right-hand sides are solved at once.

Closing note

Matrix operations are not isolated rules. Addition combines compatible datasets, multiplication composes relationships, and the matrix equation connects matrix algebra to systems of linear equations and column combinations.

Section 2.1 terminology

Term	Meaning
Matrix	Rectangular array of numbers or expressions
Size / order	Number of rows by number of columns
Entry a_{ij}	Entry in row i , column j
Row vector	Matrix with one row
Column vector	Matrix with one column
Matrix equality	Same size and equal corresponding entries
Scalar multiple	Matrix obtained by multiplying every entry by a scalar
Matrix product	Row-column multiplication when inner dimensions agree
Matrix equation	Equation such as $A\mathbf{x} = \mathbf{b}$
Partitioned matrix	Matrix divided into compatible blocks
Linear combination	Weighted sum of vectors or matrix columns

Section 2.2 - Properties of Matrix Operations

Learning objectives

Students should be able to:

- use algebraic properties of addition and scalar multiplication;
- work with zero and identity matrices;
- use associativity and distributivity of matrix multiplication;
- recognize that matrix multiplication is generally not commutative;
- solve simple matrix equations;
- compute transposes and apply transpose identities.

Algebra of matrix addition and scalar multiplication

For matrices of the same size and scalars c, d :

$$A + B = B + A, \quad A + (B + C) = (A + B) + C,$$

$$A + O = A, \quad A + (-A) = O,$$

$$(c + d)A = cA + dA, \quad c(A + B) = cA + cB,$$

$$(cd)A = c(dA), \quad 1A = A.$$

Original example: solving a matrix equation

Solve

$$3X + A = B,$$

where

$$A = \begin{bmatrix} 1 & 9 \\ 20 & 21 \end{bmatrix}, \quad B = \begin{bmatrix} 10 & 30 \\ 41 & 42 \end{bmatrix}.$$

Subtract A from both sides:

$$3X = B - A = \begin{bmatrix} 9 & 21 \\ 21 & 21 \end{bmatrix}.$$

Multiply by $\frac{1}{3}$:

$$X = \frac{1}{3}(B - A) = \begin{bmatrix} 3 & 7 \\ 7 & 7 \end{bmatrix}.$$

Properties of matrix multiplication

When sizes are compatible:

$$A(BC) = (AB)C,$$

$$A(B + C) = AB + AC, \quad (A + B)C = AC + BC,$$

$$c(AB) = (cA)B = A(cB).$$

Matrix multiplication is associative, but it is generally **not commutative**.

Detailed associativity example

Let

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 1 & 3 \end{bmatrix}, \quad C = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}.$$

First group as $(AB)C$:

$$AB = \begin{bmatrix} 5 & 2 \\ 3 & 4 \end{bmatrix},$$

so

$$(AB)C = \begin{bmatrix} 5 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 10 & 7 \\ 6 & 7 \end{bmatrix}.$$

Now group as $A(BC)$:

$$BC = \begin{bmatrix} 2 & 1 \\ 4 & 3 \\ 2 & 4 \end{bmatrix},$$

so

$$A(BC) = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 4 & 3 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} 10 & 7 \\ 6 & 7 \end{bmatrix}.$$

Both groupings give the same result.

Matrix multiplication is not commutative

Let

$$P = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad Q = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}.$$

Then

$$PQ = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}, \quad QP = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}.$$

Therefore, $PQ \neq QP$. The order of transformations matters.

Identity and zero matrices

The identity matrix I_n satisfies

$$AI_n = A, \quad I_m A = A$$

for an $m \times n$ matrix A . The zero matrix satisfies $A + O = A$. However, matrix algebra differs from ordinary number algebra: $AB = O$ does not necessarily imply $A = O$ or $B = O$.

Zero and identity matrices

The **zero matrix** O has 0 in every entry. The **identity matrix** I_n has 1s on the main diagonal and 0s elsewhere.

Matrix multiplication can have zero divisors. For example,

$$A = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix},$$

yet

$$AB = O.$$

Thus $AB = O$ does not imply $A = O$ or $B = O$.

Transpose

The transpose of $A = [a_{ij}]$ is $A^T = [a_{ji}]$: rows become columns.

If

$$M = \begin{bmatrix} 9 & 20 & 21 \\ 1 & 5 & 7 \end{bmatrix},$$

then

$$M^T = \begin{bmatrix} 9 & 1 \\ 20 & 5 \\ 21 & 7 \end{bmatrix}.$$

Important identities include

$$(A^T)^T = A, \quad (A + B)^T = A^T + B^T,$$

$$(cA)^T = cA^T, \quad (AB)^T = B^T A^T.$$

Notice the reversed order in the product identity.

Closing note

Matrix algebra resembles ordinary algebra in many ways, but multiplication has a directional structure. Associativity and distributivity survive; commutativity generally does not. The transpose also reverses multiplication order.

Section 2.2 terminology

Term	Meaning
Zero matrix O	Matrix whose entries are all zero
Additive inverse $-A$	Matrix satisfying $A + (-A) = O$
Identity matrix I_n	Square matrix with ones on the main diagonal
Associative property	Grouping does not change a valid product or sum
Distributive property	Multiplication distributes over addition
Commutative property	Order can be reversed; generally false for matrix multiplication
Transpose A^T	Matrix formed by interchanging rows and columns
Symmetric matrix	Square matrix satisfying $A^T = A$

Section 2.3 - The Inverse of a Matrix

Learning objectives

Students should be able to:

- explain invertibility and singularity;
- find a 2×2 inverse using the formula;
- find an inverse using Gauss-Jordan elimination;
- recognize when an inverse does not exist;
- use inverse properties correctly;
- solve $A\mathbf{x} = \mathbf{b}$ with A^{-1} ;
- reuse one inverse for several right-hand sides.

Definition: inverse

A square matrix A is invertible when there exists a matrix A^{-1} satisfying $AA^{-1} = A^{-1}A = I$. If no inverse exists, A is singular. An inverse, when it exists, is unique.

The 2×2 formula

For

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix},$$

A is invertible exactly when $ad - bc \neq 0$, and

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

Original example

Let

$$A = \begin{bmatrix} 3 & 2 \\ 4 & 3 \end{bmatrix}.$$

The determinant factor is

$$ad - bc = 3(3) - 2(4) = 1.$$

Therefore,

$$A^{-1} = \begin{bmatrix} 3 & -2 \\ -4 & 3 \end{bmatrix}.$$

Check:

$$AA^{-1} = \begin{bmatrix} 3 & 2 \\ 4 & 3 \end{bmatrix} \begin{bmatrix} 3 & -2 \\ -4 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Gauss-Jordan method for a 3×3 inverse

Let

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}.$$

Adjoin the identity matrix:

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{array} \right].$$

Step 1: eliminate the first entry of Row 3

Use $R_3 \leftarrow R_3 - R_1$:

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & -1 & 1 & -1 & 0 & 1 \end{array} \right].$$

Step 2: eliminate the second entry of Row 3

Use $R_3 \leftarrow R_3 + R_2$:

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 2 & -1 & 1 & 1 \end{array} \right].$$

Step 3: scale Row 3

Use $R_3 \leftarrow \frac{1}{2}R_3$:

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{array} \right].$$

Step 4: eliminate the third entry of Row 2

Use $R_2 \leftarrow R_2 - R_3$:

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & 1 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{array} \right].$$

Step 5: eliminate the second entry of Row 1

Use $R_1 \leftarrow R_1 - R_2$:

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ 0 & 1 & 0 & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & 1 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{array} \right].$$

Thus

$$A^{-1} = \frac{1}{2} \begin{bmatrix} 1 & -1 & 1 \\ 1 & 1 & -1 \\ -1 & 1 & 1 \end{bmatrix}.$$

Singular matrices

For

$$S = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix},$$

$ad - bc = 4 - 4 = 0$. The second row is twice the first, so the matrix cannot be reduced to I_2 . Therefore, S is singular.

A square matrix is **singular** precisely when it is not invertible. Equivalently,

$$A \text{ singular} \iff \det(A) = 0.$$

Properties of inverses

For invertible matrices A and B and nonzero scalar c :

$$(A^{-1})^{-1} = A, \quad (cA)^{-1} = \frac{1}{c}A^{-1},$$

$$(A^T)^{-1} = (A^{-1})^T, \quad (AB)^{-1} = B^{-1}A^{-1}.$$

The product rule reverses order.

Solving a system with an inverse

Use the same matrix

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

and solve

$$A\mathbf{x} = \begin{bmatrix} 29 \\ 41 \\ 30 \end{bmatrix}.$$

Multiply by A^{-1} :

$$\mathbf{x} = A^{-1}\mathbf{b} = \frac{1}{2} \begin{bmatrix} 1 & -1 & 1 \\ 1 & 1 & -1 \\ -1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 29 \\ 41 \\ 30 \end{bmatrix}.$$

Compute each entry:

$$\begin{aligned}
x_1 &= \frac{1}{2}(29 - 41 + 30) = 9, \\
x_2 &= \frac{1}{2}(29 + 41 - 30) = 20, \\
x_3 &= \frac{1}{2}(-29 + 41 + 30) = 21.
\end{aligned}$$

Therefore,

$$\mathbf{x} = \begin{bmatrix} 9 \\ 20 \\ 21 \end{bmatrix}.$$

Multiple right-hand sides

If the same coefficient matrix is used with many right-hand-side vectors, computing the inverse once allows all solutions to be organized in one matrix equation.

Closing note

An inverse reverses the action of an invertible matrix. The Gauss-Jordan method explains both how to find the inverse and how singularity appears when the left block cannot become the identity.

Section 2.3 terminology

Term	Meaning
Invertible / nonsingular	Matrix with a multiplicative inverse
Singular	Matrix without an inverse
Inverse A^{-1}	Matrix satisfying $AA^{-1} = A^{-1}A = I$
Adjoining I	Forming $[A \mid I]$ for Gauss-Jordan inversion
Determinant factor $ad - bc$	Test used in the 2×2 inverse formula
Inverse of a product	$(AB)^{-1} = B^{-1}A^{-1}$
Cancellation	Valid when the cancelled matrix is invertible

Section 2.4 - Elementary Matrices and LU-Factorization

Learning objectives

Students should be able to:

- construct elementary matrices;
- represent row operations by left multiplication;
- explain row equivalence using products of elementary matrices;
- factor an invertible matrix into elementary matrices;
- find an LU-factorization;
- solve $A\mathbf{x} = \mathbf{b}$ using forward and back substitution.

Bridge: elementary matrices lead to LU

Every row operation can be represented by left multiplication with an elementary matrix. If elimination gives

$$E_k \cdots E_2 E_1 A = U,$$

then reversing the steps gives

$$A = E_1^{-1} E_2^{-1} \cdots E_k^{-1} U.$$

The product of the inverse elimination matrices is lower triangular; call it L . Therefore,

$$A = LU.$$

LU factorization is Gaussian elimination with the elimination multipliers saved instead of discarded.

Definition: elementary matrix

An elementary matrix is obtained by performing one elementary row operation on an identity matrix. Left multiplication by that elementary matrix performs the same row operation on any compatible matrix.

The three types of elementary matrices

Starting with I_3 :

1. Swap R_1 and R_2 :

$$E_{\text{swap}} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

2. Scale R_2 by 5:

$$E_{\text{scale}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

3. Replace R_3 by $R_3 - 2R_1$:

$$E_{\text{replace}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}.$$

Left multiplication performs the row operation

Let

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}.$$

Then

$$E_{\text{replace}}A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 3 & 2 \end{bmatrix},$$

which is exactly the result of $R_3 \leftarrow R_3 - 2R_1$.

Factoring a matrix into elementary matrices

Let

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 7 \end{bmatrix}.$$

Reduce A to I_2 :

$$R_2 \leftarrow R_2 - 3R_1 \implies \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}.$$

Then

$$R_1 \leftarrow R_1 - 2R_2 \implies I_2.$$

The corresponding elementary matrices are

$$E_1 = \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix}, \quad E_2 = \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix}.$$

Since $E_2E_1A = I$, we have

$$A = E_1^{-1}E_2^{-1} = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}.$$

LU-factorization

An LU-factorization writes

$$A = LU,$$

where L is lower triangular and U is upper triangular.

Consider

$$A = \begin{bmatrix} 2 & 1 & -1 \\ 4 & 5 & 0 \\ -2 & 8 & 11 \end{bmatrix}.$$

Elimination step 1

Use $R_2 \leftarrow R_2 - 2R_1$. The multiplier is $m_{21} = 2$:

$$\begin{bmatrix} 2 & 1 & -1 \\ 0 & 3 & 2 \\ -2 & 8 & 11 \end{bmatrix}.$$

Elimination step 2

Use $R_3 \leftarrow R_3 + R_1 = R_3 - (-1)R_1$. The multiplier is $m_{31} = -1$:

$$\begin{bmatrix} 2 & 1 & -1 \\ 0 & 3 & 2 \\ 0 & 9 & 10 \end{bmatrix}.$$

Elimination step 3

Use $R_3 \leftarrow R_3 - 3R_2$. The multiplier is $m_{32} = 3$:

$$U = \begin{bmatrix} 2 & 1 & -1 \\ 0 & 3 & 2 \\ 0 & 0 & 4 \end{bmatrix}.$$

Store the multipliers below the diagonal:

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 3 & 1 \end{bmatrix}.$$

Therefore, $A = LU$.

The two triangular solves

Once $A = LU$,

$$A\mathbf{x} = \mathbf{b} \implies LU\mathbf{x} = \mathbf{b}.$$

Introduce $\mathbf{y} = U\mathbf{x}$. First solve

$$L\mathbf{y} = \mathbf{b}$$

by forward substitution, then solve

$$U\mathbf{x} = \mathbf{y}$$

by back substitution.

Solving a system with LU

Solve

$$A\mathbf{x} = \mathbf{b}, \quad \mathbf{b} = \begin{bmatrix} -9 \\ 49 \\ 290 \end{bmatrix}.$$

Because $A = LU$, first solve

$$L\mathbf{y} = \mathbf{b}.$$

This gives

$$\begin{cases} y_1 = -9, \\ 2y_1 + y_2 = 49, \\ -y_1 + 3y_2 + y_3 = 290. \end{cases}$$

Forward substitution:

$$y_1 = -9,$$

$$y_2 = 49 - 2(-9) = 67,$$

$$y_3 = 290 + y_1 - 3y_2 = 290 - 9 - 201 = 80.$$

Thus

$$\mathbf{y} = \begin{bmatrix} -9 \\ 67 \\ 80 \end{bmatrix}.$$

Now solve

$$U\mathbf{x} = \mathbf{y}.$$

The equations are

$$\begin{cases} 2x_1 + x_2 - x_3 = -9, \\ 3x_2 + 2x_3 = 67, \\ 4x_3 = 80. \end{cases}$$

Back substitution:

$$x_3 = 20,$$

$$3x_2 + 40 = 67 \implies x_2 = 9,$$

$$2x_1 + 9 - 20 = -9 \implies x_1 = 1.$$

Therefore,

$$\mathbf{x} = \begin{bmatrix} 1 \\ 9 \\ 20 \end{bmatrix}.$$

Why LU is useful

When many systems share the same coefficient matrix, factor $A = LU$ once. Each new right-hand side then requires only one forward substitution and one back substitution.

The main savings occurs when the same matrix A is reused with many right-hand sides: elimination is done once, and each new system needs only two triangular solves.

Closing note

Elementary matrices convert row operations into matrix multiplication. LU-factorization records the elimination process in two triangular matrices, making repeated system solving efficient and conceptually transparent.

Section 2.4 terminology

Term	Meaning
Elementary matrix	Identity matrix after one elementary row operation
Row equivalent	Related by a finite sequence of elementary row operations
Lower triangular matrix	Entries above the main diagonal are zero
Upper triangular matrix	Entries below the main diagonal are zero
LU-factorization	Factorization $A = LU$
Elimination multiplier	Number used to eliminate an entry below a pivot
Forward substitution	Solve a lower triangular system from top to bottom
Back substitution	Solve an upper triangular system from bottom to top

Section 2.5 - Markov Chains

Learning objectives

Students should be able to:

- recognize a column-stochastic matrix;
- compute state vectors using $\mathbf{x}_{k+1} = P\mathbf{x}_k$;
- compute an n th state using $P^n\mathbf{x}_0$;
- find a steady state from $P\mathbf{q} = \mathbf{q}$;
- identify regular and absorbing chains;
- interpret long-run behavior in applications.

Convention used in this course

State vectors are columns. The entry $p(i,j)$ is the probability of moving from state j to state i . Therefore, each column of a transition matrix sums to 1, and the next state is found by multiplying P by the current state.

Stochastic matrices

A column-stochastic matrix has nonnegative entries and each column sums to 1.

Original two-state R&B/pop model

Suppose listeners are classified as primarily choosing R&B or pop during a weekly listening window. Use

$$P = \begin{bmatrix} 0.80 & 0.30 \\ 0.20 & 0.70 \end{bmatrix}.$$

Column 1 describes what happens to current R&B listeners; column 2 describes what happens to current pop listeners.

Start from

$$\mathbf{x}_0 = \begin{bmatrix} 0.21 \\ 0.79 \end{bmatrix}.$$

Then

$$\mathbf{x}_1 = P\mathbf{x}_0 = \begin{bmatrix} 0.80(0.21) + 0.30(0.79) \\ 0.20(0.21) + 0.70(0.79) \end{bmatrix} = \begin{bmatrix} 0.405 \\ 0.595 \end{bmatrix}.$$

This course uses

$$p_{ij} = P(\text{next state } i \mid \text{current state } j),$$

so **columns are starting states** and **rows are next states**. The first column therefore says that a current R&B listener has an 80% chance to remain in R&B and a 20% chance to switch to pop at the next step.

Also, $\mathbf{x}_1 = P\mathbf{x}_0$ is the **next time step** after the initial distribution.

Steady state

A steady state $\mathbf{q}$ satisfies

$$P\mathbf{q} = \mathbf{q}, \quad q_1 + q_2 = 1.$$

For the music model,

$$\begin{cases} 0.8q_1 + 0.3q_2 = q_1, \\ q_1 + q_2 = 1. \end{cases}$$

The first equation gives

$$0.2q_1 = 0.3q_2 \implies 2q_1 = 3q_2.$$

Together with $q_1 + q_2 = 1$,

$$\mathbf{q} = \begin{bmatrix} 0.6 \\ 0.4 \end{bmatrix}.$$

Three-state travel-mode chain

Let the states be driving, public transit, and walking. Consider

$$P = \begin{bmatrix} 0.7 & 0.2 & 0.1 \\ 0.2 & 0.6 & 0.3 \\ 0.1 & 0.2 & 0.6 \end{bmatrix}, \quad \mathbf{x}_0 = \begin{bmatrix} 0.5 \\ 0.3 \\ 0.2 \end{bmatrix}.$$

The steady state is

$$\mathbf{q} = \begin{bmatrix} 10/29 \\ 11/29 \\ 8/29 \end{bmatrix} \approx \begin{bmatrix} 0.3448 \\ 0.3793 \\ 0.2759 \end{bmatrix}.$$

Absorbing Markov chain: hiking outcomes

Use three states:

1. still on the trail;
2. reached the summit;
3. returned to the trailhead.

Suppose

$$P = \begin{bmatrix} 0.5 & 0 & 0 \\ 0.3 & 1 & 0 \\ 0.2 & 0 & 1 \end{bmatrix}.$$

The summit and trailhead-return states are absorbing. Starting on the trail,

$$\mathbf{x}_0 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.$$

After one transition,

$$\mathbf{x}_1 = \begin{bmatrix} 0.5 \\ 0.3 \\ 0.2 \end{bmatrix}.$$

After two,

$$\mathbf{x}_2 = \begin{bmatrix} 0.25 \\ 0.45 \\ 0.30 \end{bmatrix}.$$

The probability of still being on the trail is $(1/2)^n$, so it approaches zero. The limiting absorption probabilities are

$$\lim_{n \rightarrow \infty} \mathbf{x}_n = \begin{bmatrix} 0 \\ 0.6 \\ 0.4 \end{bmatrix}.$$

Here “absorbing” means a **terminal recorded outcome**, not that a hiker is physically trapped forever. Once the experiment records “reached the summit” or “returned to the trailhead,” that trial is finished.

Closing note

Markov chains turn repeated probabilistic transitions into matrix powers. A steady state is a balance vector that is unchanged by the transition matrix. Absorbing chains model eventual outcomes rather than long-run mixing.

Section 2.5 terminology

Term	Meaning
State	Category occupied by a member of the population
Transition probability	Probability of moving from one state to another
Stochastic matrix	Nonnegative matrix whose columns sum to 1
State vector	Column of state proportions or probabilities
Markov chain	Repeated process governed by a fixed transition matrix
n th state	$\mathbf{x}_n = P^n \mathbf{x}_0$
Steady state	Probability vector satisfying $P\mathbf{q} = \mathbf{q}$
Regular matrix	Stochastic matrix with some positive power
Absorbing state	State with probability 1 of remaining there
Absorbing chain	Chain that can eventually reach an absorbing state

Section 2.6 - More Applications of Matrix Operations

Learning objectives

Students should be able to:

- encode and decode a message using an invertible matrix;
- formulate and solve an open Leontief input-output model;
- derive the normal equations for least squares regression;
- compute and interpret a least squares line and residual error.

Matrix cryptography

Use the mapping

$$\text{blank} = 0, \quad A = 1, \quad B = 2, \quad \dots, \quad Z = 26.$$

Encode row vectors on the right using

$$E = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix}.$$

Because

$$\det(E) = 1(3) - 2(1) = 1,$$

E is invertible and

$$E^{-1} = \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix}.$$

Original message: GO SPURS GO

The numerical sequence is

$$7, 15, 0, 19, 16, 21, 18, 19, 0, 7, 15, 0.$$

Partition into pairs:

$$[7 \ 15], [0 \ 19], [16 \ 21], [18 \ 19], [0 \ 7], [15 \ 0].$$

Encode each pair:

$$[7 \ 15]E = [22 \ 59],$$

$$[0 \ 19]E = [19 \ 57],$$

$$[16 \ 21]E = [37 \ 95],$$

$$[18 \ 19]E = [37 \ 93],$$

$$[0 \ 7]E = [7 \ 21],$$

$$[15 \ 0]E = [15 \ 30].$$

The cryptogram is

$$22, 59, 19, 57, 37, 95, 37, 93, 7, 21, 15, 30.$$

To decode, multiply each coded row by E^{-1} on the right.

Leontief input-output model

Consider three fictional industries:

1. travel services;
2. music and entertainment;
3. sporting goods.

Let

$$D = \begin{bmatrix} 0.10 & 0.05 & 0.10 \\ 0.20 & 0.10 & 0.05 \\ 0.05 & 0.10 & 0.10 \end{bmatrix}$$

be the internal consumption matrix. If $\mathbf{x}$ is total output and $\mathbf{e}$ is external demand, then

$$\mathbf{x} = D\mathbf{x} + \mathbf{e}.$$

Therefore,

$$(I - D)\mathbf{x} = \mathbf{e}.$$

Suppose

$$\mathbf{e} = \begin{bmatrix} 95 \\ 62 \\ 56 \end{bmatrix}.$$

Then

$$I - D = \begin{bmatrix} 0.90 & -0.05 & -0.10 \\ -0.20 & 0.90 & -0.05 \\ -0.05 & -0.10 & 0.90 \end{bmatrix}.$$

Multiplying by 20 produces an equivalent integer system:

$$\begin{bmatrix} 18 & -1 & -2 \\ -4 & 18 & -1 \\ -1 & -2 & 18 \end{bmatrix} \mathbf{x} = \begin{bmatrix} 1900 \\ 1240 \\ 1120 \end{bmatrix}.$$

Solving gives

$$\mathbf{x} = \begin{bmatrix} 120 \\ 100 \\ 80 \end{bmatrix}.$$

Verification:

$$D\mathbf{x} + \mathbf{e} = \begin{bmatrix} 25 \\ 38 \\ 24 \end{bmatrix} + \begin{bmatrix} 95 \\ 62 \\ 56 \end{bmatrix} = \begin{bmatrix} 120 \\ 100 \\ 80 \end{bmatrix}.$$

The scaled system can also be shown as

$$\left[\begin{array}{ccc|c} 18 & -1 & -2 & 1900 \\ -4 & 18 & -1 & 1240 \\ -1 & -2 & 18 & 1120 \end{array} \right].$$

For the first industry,

$$0.10(120) + 0.05(100) + 0.10(80) = 12 + 5 + 8 = 25$$

units are used internally, and

$$25 + 95 = 120$$

verifies the first total-output component.

Least squares regression

Suppose an original travel-related dataset is

$$(1, 8), (2, 9), (3, 17), (4, 17), (5, 24).$$

Fit

$$y = a_0 + a_1x.$$

Write

$$X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \\ 1 & 5 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 8 \\ 9 \\ 17 \\ 17 \\ 24 \end{bmatrix}.$$

The normal equations are

$$X^T X \mathbf{a} = X^T \mathbf{y}.$$

Compute

$$X^T X = \begin{bmatrix} 5 & 15 \\ 15 & 55 \end{bmatrix}, \quad X^T \mathbf{y} = \begin{bmatrix} 75 \\ 265 \end{bmatrix}.$$

Thus

$$\begin{cases} 5a_0 + 15a_1 = 75, \\ 15a_0 + 55a_1 = 265. \end{cases}$$

Multiply the first equation by 3:

$$15a_0 + 45a_1 = 225.$$

Subtract from the second equation:

$$10a_1 = 40, \quad a_1 = 4.$$

Then

$$5a_0 + 15(4) = 75 \implies a_0 = 3.$$

The least squares line is

$$\hat{y} = 3 + 4x.$$

Predicted values are 7, 11, 15, 19, 23, so the residual vector is

$$\mathbf{r} = \begin{bmatrix} 1 \\ -2 \\ 2 \\ -2 \\ 1 \end{bmatrix}.$$

The sum of squared errors is

$$\text{SSE} = 1^2 + (-2)^2 + 2^2 + (-2)^2 + 1^2 = 14.$$

Closing note

Cryptography uses invertible matrix transformations, Leontief models use $(I - D)^{-1}$ to account for internal demand, and least squares uses matrix projection ideas to fit inconsistent data. These applications show how the same matrix operations support very different fields.

Section 2.6 terminology

Term	Meaning
Cryptogram	Encoded numerical message
Encoding matrix	Invertible matrix used to transform message blocks
Decoding matrix	Inverse of the encoding matrix

Term	Meaning
Input-output matrix D	Internal consumption coefficients in a Leontief model
External demand $\mathbf{e}$	Demand from outside the modeled industries
Total output $\mathbf{x}$	Production required to meet internal and external demand
Least squares line	Line minimizing the sum of squared residuals
Design matrix X	Matrix containing regression predictor structure
Normal equations	$X^T X \mathbf{a} = X^T \mathbf{y}$
Residual	Observed value minus predicted value
SSE	Sum of squared errors

Chapter 2 Summary

The chapter develops a connected sequence:

matrix operations $\longrightarrow$ matrix algebra $\longrightarrow$ inverses and elementary matrices $\longrightarrow$ LU-factorization $\longrightarrow$ dynamic

Core computational habits:

1. check matrix dimensions before every operation;
2. show row-column products explicitly when multiplying;
3. verify inverses by multiplication;
4. record every elementary row operation;
5. keep the order of matrix products intact;
6. interpret matrix results in the original application context;
7. use Python to verify, not replace, mathematical reasoning.

Additional clarification: multiple right-hand sides

Suppose the same invertible coefficient matrix A is used with several right-hand sides:

$$A\mathbf{x}_1 = \mathbf{b}_1, \quad A\mathbf{x}_2 = \mathbf{b}_2, \quad \dots, \quad A\mathbf{x}_k = \mathbf{b}_k.$$

Collect the unknown vectors and right-hand sides into matrices:

$$X = [\mathbf{x}_1 \quad \mathbf{x}_2 \quad \dots \quad \mathbf{x}_k], \quad B = [\mathbf{b}_1 \quad \mathbf{b}_2 \quad \dots \quad \mathbf{b}_k].$$

Then all systems can be written at once as

$$AX = B.$$

If A^{-1} is already available, then

$$X = A^{-1}B.$$

The j th column of X is the solution corresponding to the j th right-hand side. In practice, LU factorization is usually preferable to explicitly forming A^{-1} for repeated numerical solves, but this matrix equation explains the underlying structure.